

CSS Flexbox Layouts and Card Components



Objective 01

What is CSS and Flexbox?

Objective 02

Container vs Item Concept

Objective 03

Flexbox Properties & Card Components



Learning Objectives

What We'll Discover Today!

Get ready to learn amazing skills that real web developers use every day. By the end of this session, you'll be able to create beautiful card layouts just like the ones you see on your favorite apps!



Why Flexbox Matters

Which website would you use?

Without Flexbox

Messy unstyled webpage
with stacked text blocks

With Flexbox

Clean card-based layout
with neat boxes like Zomato
restaurant cards





What is CSS?

CSS Adds Colors, Styles, and Layouts to HTML

Think of HTML as the skeleton of a webpage, and CSS as the decoration! Just like a plain white kurta becomes a beautiful festival outfit with colorful embroidery and patterns, CSS transforms plain HTML into something amazing!

Tiffin Box

The tiffin box (dabba) is the Flex Container – it holds everything together

Compartments

Each compartment (rice, dal, vegetables) is a Flex Item – neatly organized inside

Auto Arrange

Flexbox automatically fits items perfectly, just like dabba compartments!



What is Flexbox?

A Magic Organizer for Your Webpage!

Flexbox arranges elements automatically in rows or columns – like a magic shelf that organizes items for you without any effort!

Container

The Parent Element – Like a teacher giving orders to the class

Items

The Children Elements – Like students who follow the teacher's instructions

Assembly

Teacher says "Stand in a line!" and students obey – that's Flexbox!

Container vs Item

Understanding Parent and Child Elements in Flexbox

Think of it like school assembly! The teacher (container) gives instructions, and students (items) follow. The container is the boss that controls how items are arranged.



display: flex - The ON Switch

Turn on Flexbox to arrange items in a row!

Before: No Flexbox

Items stack vertically like a pile of books on the floor

After: display: flex

Items line up horizontally in a neat row, like books on a shelf!



flex-start

Items crowd to the LEFT side, like food pushed to one corner of the Meal platter

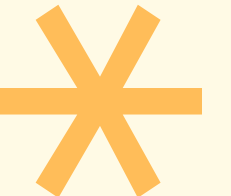
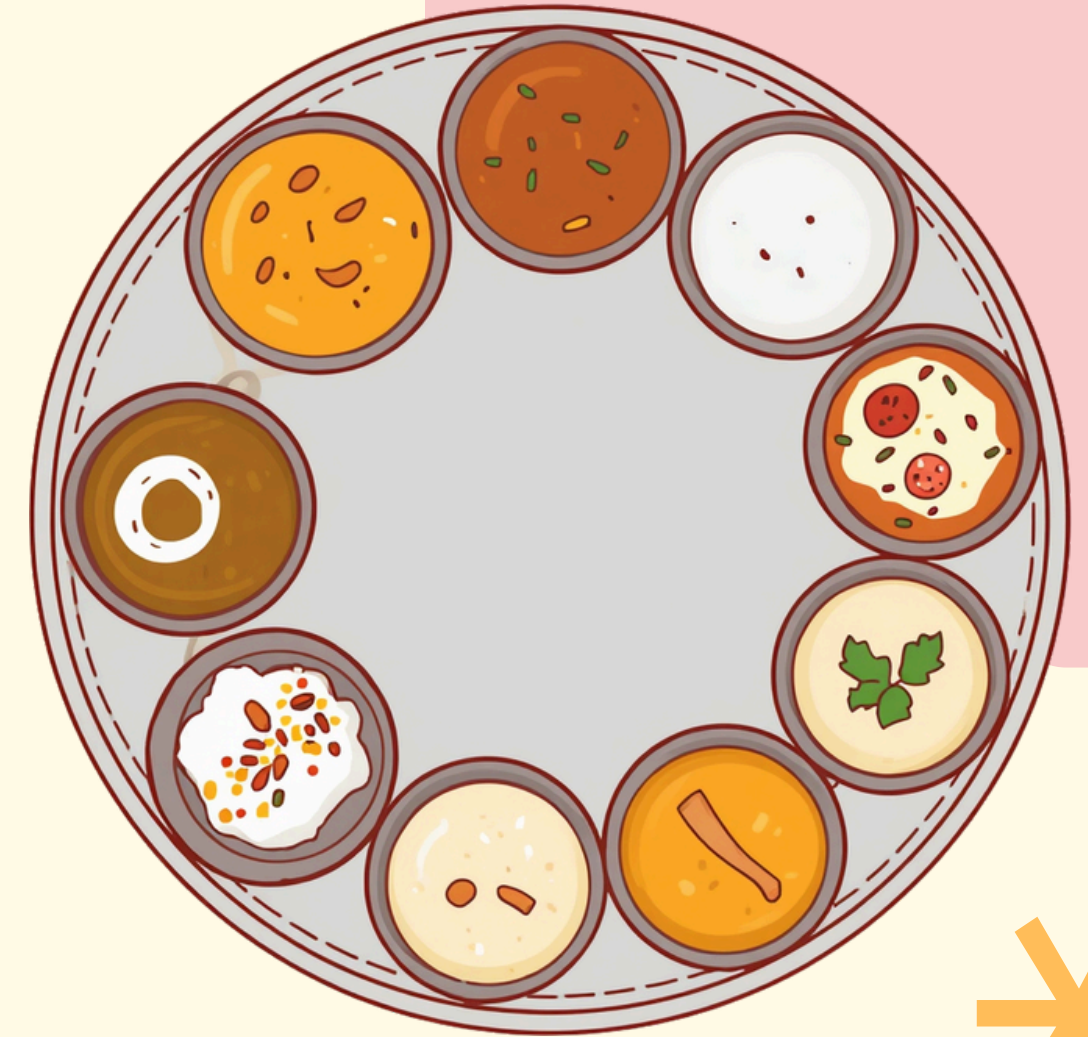


center

Items sit in the MIDDLE, like dal bowl placed at the center of your Meal platter

space-between

Equal space BETWEEN items, like "Small bowls" spread evenly across the Meal platter



justify-content Property

Controls LEFT-RIGHT Spacing on the Main Axis

Think of a thali plate! Just like you arrange roti, dal, and "Vegetable" on your plate, justify-content arranges items left-to-right in your flex container.



flex-start

Items align at the TOP, like books placed on the top shelf

center

Items align in the MIDDLE vertically, perfectly balanced

flex-end

Items align at the BOTTOM, like books resting on the floor



align-items Property

Controls UP-DOWN Alignment on the Cross Axis

Think of a bookshelf! Where do your books sit vertically? At the top, middle, bottom, or stretched to fill the whole shelf? That's exactly what align-items does for your flex items!



flex-direction Property

Sets the Main Axis Direction for Your Layout

column

Vertical top to bottom, like floors stacked in a building

row (default)

Horizontal left to right, like students sitting in classroom seats



Zomato Cards

Restaurant photo + name
+ rating + delivery time

Flipkart Cards

Product image + name +
price + Add to Cart button

BYJU'S Cards

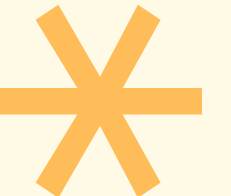
Subject image + class name
+ teacher + enroll button



What is a Card Component?

Self-Contained Boxes That Display Information Beautifully

A card is like a small poster with an image, title, and description all in one neat box. You see them everywhere on your favorite apps!



Step 1

Create a div with class "card" as the container

Step 2

Add card-image div for the picture area

Step 3

Add h3 for title and p for description



Build Step 1: HTML Structure

Creating the Card Skeleton Code

Just like building blocks, we stack our HTML elements to create a card. Each part has a special job!

* Build Step 2: Card Base CSS

display: flex

Turns ON Flexbox – items now follow our arrangement rules!

flex-direction: column

Stacks items vertically like floors in a building!

border-radius: 12px

Makes corners round and friendly – like a greeting card!



Styling the Card Container with Flexbox Properties

Now we add CSS to make our card look nice! We use `display: flex` to turn on Flexbox magic, `flex-direction: column` to stack items top-to-bottom, and `align-items: center` to keep everything in the middle.

Step 01

width: 100% makes the image fill the full card width

Step 02

height: 120px sets a fixed height for the image box

Step 03

background-color: #FFA500 gives it a bright orange color!



Build Step 3: Image Placeholder

Styling the Card Image Area

The image area is like a picture frame inside your card – it holds the visual that catches everyone's eye first!



Title Style

font-size: 16px makes title readable,
color: #333 for dark text

Description

font-size: 13px for smaller text, color:
#777 for softer gray look

Alignment

text-align: center keeps description
neat and balanced in card



Build Step 4 - Text Styling

Making Your Card Title and Description Look Great!

Now we style the text inside our card using
font-size for how big, color for what shade,
and text-align to center our description
nicely.

Step 1

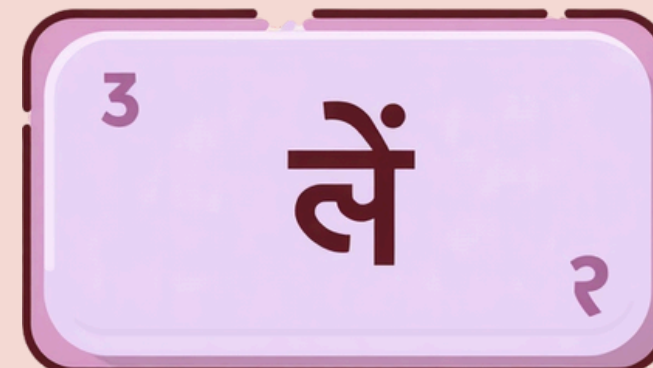
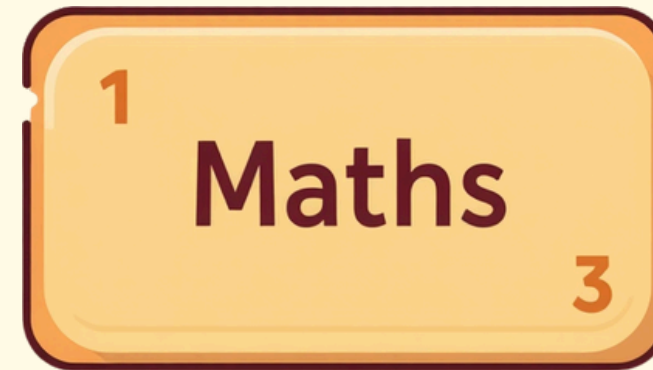
Create a container div with class "card-row" around all cards

Step 2

Add display: flex and flex-direction: row to line up cards

Step 3

Use justify-content: space-between and gap: 16px for spacing



Build Step 5 - Multiple Cards in Row

Creating a Row of Cards Using Flexbox

Wrap your cards in a container div and use display: flex to arrange them horizontally with neat spacing!

display: flex

The ON Switch!
Turns on Flexbox magic

justify-content

LEFT-RIGHT Spacing
Like arranging food on thali

align-items

UP-DOWN Alignment
Like books on a shelf



Recap Flash Cards

Quick Review of Key Flexbox Concepts

Let's remember the magic tools we learned today! Each flashcard shows a superpower that helps us arrange things beautifully on our webpage.





Homework

Find 3 cards on Zomato, Flipkart, or any app. Screenshot or draw them!

Next Session

Make cards INTERACTIVE with hover effects and mobile-friendly design!

Challenge

What is inside each card you found? Write down the elements!



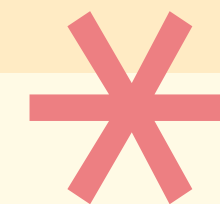
Homework and Next Session

You Built Your First Flexbox Layout!

That's what real web developers do! You're on your way to becoming a coding superstar. Keep practicing and exploring!



Thank You!



Any Questions? We'd Love to Hear From You!

